



SCHOOL OF COMPUTER SCIENCE

APPLIED MACHINE LEARNING PROJECT

PREDICTING THE RESULTS OF PROFESSIONAL TENNIS MATCHES WITH MACHINE LEARNING

Author: Sergio de Carvalho, D24130831

Date: 31/05/2026

Declaration

I hereby declare that the work described in this report is, except where otherwise stated, entirely my own and that I have not submitted it before for summative assessment in third-level education. I have read and understand TU Dublin's policy on plagiarism.

Signed[*Joe Bloggs*]..... Date[31/05/2026].....

(Sergio de Carvalho, D24130831)

Acknowledgements

[Please leave this section out if you do not have any acknowledgements.]

Abstract

[This should be up to 200 words and provide a summary of what was done and important findings.]

Contents

1	Introduction	7
1.1	Project description	7
1.2	Project motivation and justification	7
1.3	Project scope	7
1.4	Report layout	8
2	Literature review	9
3	Methodology	10
4	Results	11
5	Conclusion	12
	References	13
A	Examples of figure, table and equation	14

Acronyms and Abbreviations

[	
ATP	Association of Tennis Professionals
WTA	Women's Tennis Association
]	

1 Introduction

Tennis is a popular sport where the one-on-one nature of competition provides an ideal environment for sports analytics, as match outcomes are determined by a more controlled set of variables than team-based sports. While professional tennis has benefited from rigorous record-keeping and a wealth of statistics about every aspect of the game, accurately predicting match results remains a complex challenge due to the intricate interactions between player skills, form, physical and psychological state, as well as environmental factors such as weather, court surface type, and tennis ball characteristics that can significantly influence the performance of players and the outcome of a match.

1.1 Project description

The focus of this project is to develop a framework that prioritises the curation of high-quality data and the engineering of domain-driven features that can capture the multiple facets of the game and be used to predict the results of professional tennis matches. Central to this approach is the use of “enhanced covariates” — variables that are not directly observed but are derived through mathematical transformations or separate statistical models.

There has been significant research on developing statistical models to rank tennis players as alternatives to the official rankings published by the professional tennis organisations. In parallel, several studies have applied machine learning to predict the results of tennis matches, however, most of these studies were restricted to specific tournaments (e.g. Grand Slam tournaments only) or did not take full advantage of the available data and statistical models. This project aims to address these limitations by tapping into the vast amount of data available, and use machine learning techniques to explore the predictive power of different statistical models proposed in the literature while also discovering new ways to improve the predictions of these models.

1.2 Project motivation and justification

Predicting the outcomes of sporting events has been a long-standing problem in the field of sports analytics, and can have significant implications for the development of training methods and strategies to improve the performance of professional athletes (Almarashi, Daniyal, and Jamal 2024). It is also important for broadcasters and the media in general as it helps them to transform a passive viewing experience into an interactive, high-stakes narrative that drives viewer retention and fan engagement. Finally, it can be argued that predicting sporting events can help develop valuable models and gain insights that could be helpful in diverse areas of human activity, from financial markets, to politics and international conflicts (Gu and Saaty 2019).

1.3 Project scope

This project is focused on predicting the results of professional tennis matches on the main events organised by the Association of Tennis Professionals (ATP), the men’s tennis circuit. While there is a wealth of data available from the Women’s Tennis Association (WTA) as well, limiting the scope to one of the two circuits

allows for a more focused and comprehensive analysis in the limited time available for this project.

We note that there has been a significant interest in the literature in using betting odds to either improve the predictions or measure their accuracy. While odds carefully curated and published by professional bookmakers can be a valuable tool of information, this project does not use odds data nor focuses on this aspect.

1.4 Report layout

[A sentence on each of the following sections, introducing the reader to the report.]

2 Literature review

See Buhamra, Groll, and Gerharz (2025) for the details of the modern machine learning approaches and different forecast strategies on Grand Slam tennis tournaments.

See Buhamra and Groll (2025) for the details of the statistical enhanced learning for modeling and prediction tennis matches at Grand Slam tournaments.

See Trucco and Tolić (2025) for the details of the machine learning framework for forecasting professional tennis matches using ATP statistics and betting odds.

See Almarashi, Daniyal, and Jamal (2024) for the details of the novel comparative study of NNAR approach with linear stochastic time series models in predicting tennis player's performance.

See Buhamra, Groll, and Brunner (2024) for the details of the modeling and prediction of tennis matches at Grand Slam tournaments.

See Fayomi et al. (2022) for the details of the forecasting tennis match results using the Bradley-Terry model.

See Yue et al. (2022) for the details of the study of forecasting tennis matches via the Glicko model.

See Angelini, Candila, and De Angelis (2022) for the details of the Weighted Elo rating method.

See Williams et al. (2021) for the details of the how well do Elo-based ratings predict professional tennis matches.

See Wilkens (2021) for the details of the sports prediction and betting models in the machine learning age: The case of tennis.

See Gao and Kowalczyk (2021) for the details of the random forest model identifies serve strength as a key predictor of tennis match outcome.

See Gu and Saaty (2019) for the details of the predicting the outcome of a tennis tournament: Based on both data and judgments.

See Knottenbelt, Spanias, and Madurska (2012) for the details of the a common-opponent stochastic model for predicting the outcome of professional tennis matches.

See McHale and Morton (2011) for the details of the A Bradley-Terry type model for forecasting tennis match results.

See Somboonphokkaphan, Phimoltares, and Lursinsap (2009) for the details of the tennis winner prediction based on time-series history with neural modeling.

3 Methodology

[You may structure this as you wish, to suit your particular project. Include information on the stages of the project and details of how each stage was carried out. In addition, explain and justify (either from common knowledge in the field, with reference to sources or through logical deduction) the choice of approach at all levels. A good rule of thumb for checking whether the argumentation in this (or any other) section is ‘watertight’ is to ask yourself if there is anything in the text that the reader could reasonably ask ‘why’ or ‘how can you be sure?’ about without getting an answer in the text itself – if there is, then you have more writing to do.

Examples of project stages are data collection and preparation, modelling, model selection and tuning, model evaluation and data interpretation (this is not an exhaustive list). However, do not include each and every type of task only because you learnt about in the course. Many will not be applicable in your project (e.g. data cleaning, scaling or PCA will not always be needed). If you are ‘skipping’ a common task such as data cleaning, just explain why, as this one of the decisions you made with regard to the methodology.

Discuss the implications of the methodology in the application domain.

Information on tools (e.g. Colab, Python libraries etc.) should be presented in a subsection of this chapter and not ‘woven’ through the conceptual account of the methodology.]

4 Results

[You may structure this as you wish, but let the reader know everything you know about the results, regardless of whether you consider the outcome good or bad.

Discuss the results, rather than simply presenting them to your reader and letting them do the work.

Be inquisitive when analysing your results. For example, if you are getting accuracies in the high 90s there is probably either an extreme imbalance in the data set or a data leak. Always investigate and at least discuss but, if there is time, it is best to redesign and redo your experiments.

Discuss the implication of the results in the application domain]

5 Conclusion

[Structure as you wish but make sure that it includes a synthesis of the results, a high-level discussion of their implications in the application domain, any particularly interesting low-level details of the results and, finally, some words on future work i.e. how you or someone else might build on the project.]

References

- Almarashi, A.M., M. Daniyal, and F. Jamal (2024). "A novel comparative study of NNAR approach with linear stochastic time series models in predicting tennis player's performance". In: *BMC Sports Science, Medicine & Rehabilitation* 16.1, pp. 28–11.
- Angelini, G., V. Candila, and L. De Angelis (2022). "Weighted Elo rating for tennis match predictions". In: *European Journal of Operational Research* 297.1, pp. 120–132. ISSN: 0377-2217. DOI: <https://doi.org/10.1016/j.ejor.2021.04.011>. URL: <https://www.sciencedirect.com/science/article/pii/S0377221721003234>.
- Buhamra, N. and A. Groll (2025). *Statistical enhanced learning for modeling and prediction tennis matches at Grand Slam tournaments*. Tech. rep. Cornell University Library.
- Buhamra, N., A. Groll, and S. Brunner (2024). "Modeling and prediction of tennis matches at Grand Slam tournaments". In: *Journal of Sports Analytics* 10.1, pp. 17–33.
- Buhamra, N., A. Groll, and A. Gerharz (2025). "Comparing modern machine learning approaches and different forecast strategies on Grand Slam tennis tournaments". In: *Journal of Sports Analytics* 11.
- Fayomi, A. et al. (2022). "Forecasting tennis match results using the Bradley-Terry model". In: *International Journal of Photoenergy* 2022, pp. 1–12.
- Gao, Z. and A. Kowalczyk (2021). "Random forest model identifies serve strength as a key predictor of tennis match outcome". In: *Journal of Sports Analytics* 7.4, pp. 255–262.
- Gu, W. and T.L. Saaty (2019). "Predicting the outcome of a tennis tournament: Based on both data and judgments". In: *Journal of Systems Science and Systems Engineering* 28.3, pp. 317–343.
- Knottenbelt, W.J., D. Spanias, and A.M. Madurska (2012). "A common-opponent stochastic model for predicting the outcome of professional tennis matches". In: *Computers & Mathematics with Applications* 64.12, pp. 3820–3827.
- McHale, I. and A. Morton (2011). "A Bradley-Terry type model for forecasting tennis match results". In: *International Journal of Forecasting* 27.2, pp. 619–630. ISSN: 0169-2070. DOI: <https://doi.org/10.1016/j.ijforecast.2010.04.004>. URL: <https://www.sciencedirect.com/science/article/pii/S0169207010001019>.
- Somboonphokkaphan, A., S. Phimoltares, and C. Lursinsap (2009). "Tennis winner prediction based on time-series history with neural modeling". In: *Proceedings of the International Multiconference of Engineers and Computer Scientists*. Vol. 1.
- Trucco, L. and D. Tolić (2025). "A framework for forecasting professional tennis matches using ATP statistics and betting odds". In: *2025 10th International Conference on Machine Learning Technologies (ICMLT)*, pp. 292–297. DOI: 10.1109/ICMLT65785.2025.11193322.
- Wilkens, S. (2021). "Sports prediction and betting models in the machine learning age: The case of tennis". In: *Journal of Sports Analytics* 7.2, pp. 99–117.
- Williams, L.V. et al. (2021). "How well do Elo-based ratings predict professional tennis matches?" In: *Journal of Quantitative Analysis in Sports* 17.2, pp. 91–105.
- Yue, Jack C. et al. (2022). "A study of forecasting tennis matches via the Glicko model". In: *PLoS One* 17.4, e0266838.

A Examples of figure, table and equation

Figure example

Example of figure:

Figure 1: Example caption

And here is an example of a reference to Figure 1.

Table example

Example of table:

Table 1: Correlation values for N=100 and $\alpha = .05$

Pair	r	Significant
First	0.22	Yes
Second	0.07	No
Third	0.31	Yes

An example of referencing Table 1.

Equation example

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{(n-1)S_x S_y} = \frac{S_{xy}^2}{S_x S_y} \quad (1)$$

where x, y are the two variables, x_i, y_i the i^{th} values, $\bar{x}, \bar{y}$ the means, S_x and S_y the sample standard deviations and S_{xy}^2 the sample covariance. And here is a reference to Equation 1.]